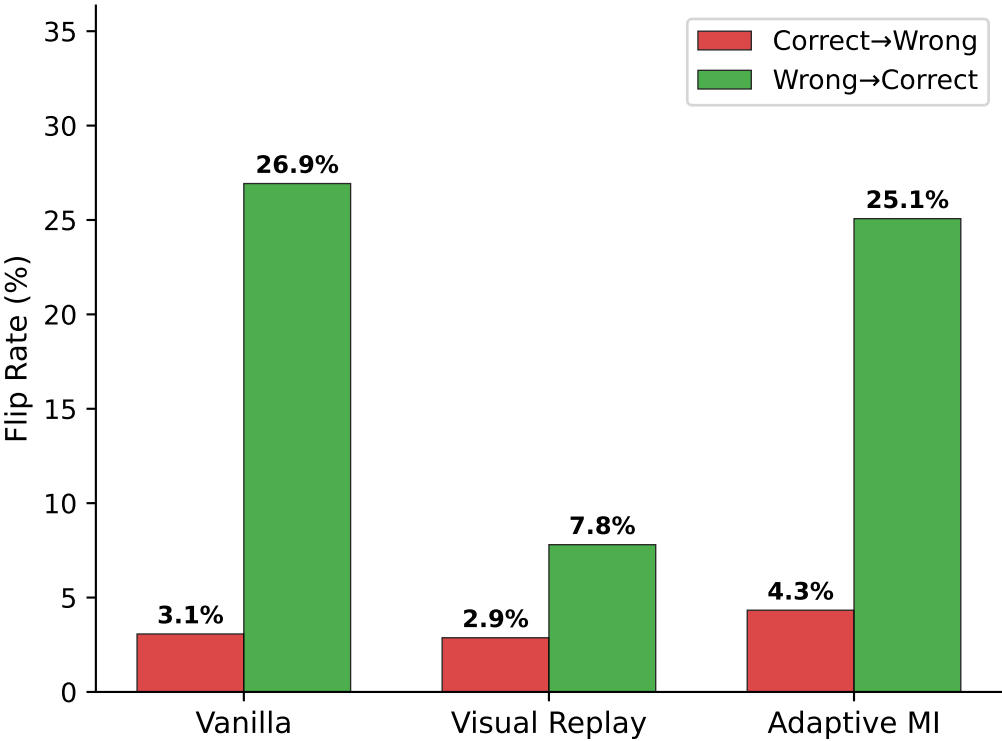
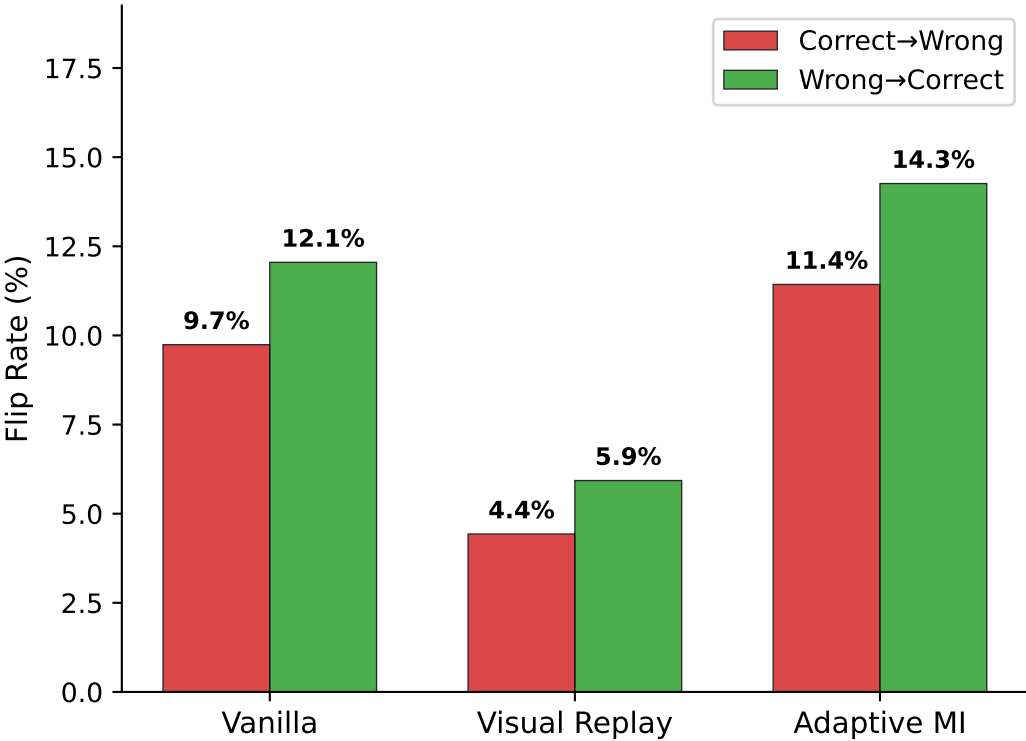


Budget Flip Rate Analysis: Short (128) vs Long (512) Tokens

Flip Rates on MMSTAR (128→512 tokens)



Flip Rates on HALLUSIONBENCH (128→512 tokens)



MMStar Flip Rates by Category (128→512 tokens)

Category	Vanilla C→W%	Vanilla W→C%	Vanilla Net%	Visual Replay C→W%	Visual Replay W→C%	Visual Replay Net%	Adaptive MI C→W%	Adaptive MI W→C%	Adaptive MI Net%
coarse perception	2.0	21.2	-19.2	1.2	2.4	-1.2	2.0	21.2	-19.2
fine-grained perception	2.4	10.8	-8.4	1.6	1.6	0.0	3.6	16.4	-12.8
	2.4	28.0	-25.6	2.0	4.8	-2.8	2.4	28.0	-25.6
logical reasoning	2.0	35.2	-33.2	2.4	10.4	-8.0	4.8	28.0	-23.2
math	2.4	42.8	-40.4	5.2	18.8	-13.6	3.2	33.2	-30.0
science & technology	7.2	23.6	-16.4	4.8	8.8	-4.0	10.0	23.6	-13.6